2.2 Experiment 2 – System identification

In this experiment, the parameters of various dynamic systems have to be determined experimentally. For this purpose, step responses or impulse responses (see $x_e(t)$) were recorded and saved as text files. All parameters (numerical values and units) necessary for a complete description of the system must be determined using Matlab or Python.
Document every step thoroughly; show the graphs of the system responses with correct axis labels and add the evaluations you have programmed (tangents, intersections, etc.) and explain your approach clearly and concisely! Display

a) **V→** Save the data provided on ELO in a directory on a local drive. Work through the complete experiment description. The basic knowledge of control engineering required for this experiment has to be reviewed in the RTV lecture or the lecture notes. This knowledge is tested by asking questions at the beginning of the experiment.

b) **V→** Explain at least five different basic system behaviors!

c) **V→** What parameters are needed to represent the above systems?

d) **V→** Sketch the step and impulse responses of at least two of the above systems and specify the differential equation and transfer function that describe the respective system!

e) Start Matlab/Python and select a directory on your local drive as the working directory. Please avoid spaces, special characters or umlauts in the working directory path and all file names! Work only in this directory! All operations must be performed using Matlab or Python. The use of other software (e.g. Excel) is not permitted. Show the commands that are essential for importing the data.

f) Determine the system behavior experimentally using the recorded step or impulse responses of the given systems and their parameters (values and units!). Document the results, show the essential commands and specify the differential equation and transfer function describing the system.

g) Provide a real example for each of the determined system behaviors with a description of the respective input and output variables!

h) Compare the given system responses with the responses of your identified system in the same graphs and also show the deviation for each system!

Note: Useful commands: `cumsum`, `diff`, `polyfit`, `polyval`.